Gas Laws

1. The temperature of a gas at pressure P and volume V is $27°C$. Keeping its volume constant if its temperature is raised to $927°C$, then its pressure will be

(a) $2P$ (b) $3P$

(c) $4P$ (d) $6P$

2. 4 moles of an ideal gas is at $0°C$. At constant pressure it is heated to double its volume, then its final temperature will be

(a) $0°C$ (b) $273°C$

(c) $546°C$ (d) $136.5°C$

3. Every gas (real gas) behaves as an ideal gas

(a) At high temperature and low pressure

(b) At low temperature and high pressure

(c) At normal temperature and pressure

(d) None of the above

4. Boyle's law holds for an ideal gas during

(a) Isobaric changes (b) Isothermal changes

(c) Isochoric changes (d) Isotonic changes

5. S.I. unit of universal gas constant is

(a) $cal/°C$ (b) J/mol

(c) $Jmol^{-1}K^{-1}$ (d) J/kg

6. Molecules of a gas behave like

(a) Inelastic rigid sphere

(b) Perfectly elastic non-rigid sphere

(c) Perfectly elastic rigid sphere

(d) Inelastic non-rigid sphere

7. A perfect gas at $27°C$ is heated at constant pressure so as to triple its volume. The temperature of the gas will be

(a) $81°C$ (b) $900°C$

(c) $627°C$ (d) $450°C$

8. The density of a gas at normal pressure and $27°C$ temperature is 24. Keeping the pressure constant, the density at $127°C$ will be

(a) 6 (b) 12

(c) 18 (d) 24

9. The volume of a gas at 20°*C* is 200 *ml*. If the temperature is reduced to – 20°*C* at constant pressure, its volume will be

(a) 172.6 *ml* (b) 17.26 *ml*

(c) 192.7 *ml* (d) 19.27 *ml*

10. Two gases of equal mass are in thermal equilibrium. If P_a, P_b and V_a and V_b are their respective pressures and volumes, then which relation is true

(a) $P_a \neq P_b; V_a = V_b$

(b) $P_a = P_b; V_a \neq V_b$

(c) $\frac{P_a}{V_a} = \frac{P_b}{V_b}$

(d) $P_a V_a = P_b V_b$

11. At absolute zero temperature, pressure of a gas will be

(a) Zero

(b) One atmospheric pressure

(c) $P_0 \times 273$

(d) $P_0 \times 76$

12. The gas which obeys Boyle's law for maximum range of temperature is

(a) CO_2 (b) O_3

(c) H_2 (d) *He*

13. The vapour of a substance behaves as a gas

(a) Below critical temperature

(b) Above critical temperature

(c) At 100°*C*

(d) At 1000°*C*

14. The temperature below which a gas should be cooled, before it can be liquified by pressure only is termed as

(a) The dew point

(b) The freezing point

(c) The saturation point

(d) The critical point

15. It is possible for a substance to coexist in all three phases in equilibrium, when the substance is at

(a) Boyle temperature

(b) Critical temperature

(c) Triple point

(d) Dew point

16. When air is filled in the balloon, the pressure and volume both increases while temperature does not change.

Here Boyle's law is not obeyed because

(a) Mass of air is negligible

(b) Mass of air does not remain constant

(c) Air is not perfect gas

(d) Pressure inside the balloon is less than that of the atmospheric pressure

17. The mean free path of gas molecules depends on (d = molecular diameter)

(a) d (b) d^2

(c) d^{-2} (d) d^{-1}

18. For Boyle's law to hold the gas should be

(a) Perfect and of constant mass and temperature

(b) Real and of constant mass and temperature

(c) Perfect and at constant temperature but variable mass

(d) Real and at constant temperature but variable mass

19. If pressure of a gas contained in a closed vessel is increased by 0.4% when heated by 1°C, the initial temperature must be

(a) 250 K (b) 250°C

(c) 2500 K (d) 25°C

20. Consider a 1 *c.c.* sample of air at absolute temperature T_0 at sea level and another 1 *cc* sample of air at a height where the pressure is one-third atmosphere. The absolute temperature T of the sample at that height is

(a) Equal to $T_0/3$

(b) Equal to $3/T_0$

(c) Equal to T_0

(d) Cannot be determined in terms of T_0 from the above data